

China emerging as a global biotechnology player

Date: 2026-01-28

Source: <https://www.franklintempleton.com/articles/2025/clearbridge-investments/china-emerging-as-a-global-biotechnology-player>

Extraction method: selenium

China emerging as a global biotechnology player

Key takeaways

Significant state-supported investment in research and development, combined with established life science infrastructure, is bolstering the rapid growth of a Chinese biotechnology industry.

China has seen early success as "fast followers," licensing and commercializing drugs developed by Western biopharmaceutical companies into a domestic health care market with large patient populations.

We believe China must develop and commercialize its own treatments to become a profitable biopharmaceutical competitor. This has begun to occur as an expanding number of Chinese biotechs are now licensing new compounds to multinational pharmaceutical firms.

Swift evolution sparked by healthcare push, R&D spending

China's quest for independence from the West, most notably the rise of its own technological prowess as evidenced by global leadership in electric vehicle production and the success of its DeepSeek AI model, is also reshaping an overlooked part of the world's second-largest economy: health care. Just five years ago, China was dependent on Western companies for access to leading drugs. Today it is investing heavily in the research and development (R&D) necessary to create its own therapeutics through a homegrown biotechnology industry. The fruits of these innovative efforts are starting to show, which we believe will create opportunities for global investors.

A viable biotechnology industry is part of the country's push over the last 50 years to elevate its health care system to treat an aging and sick Chinese population. Commercial biotech treatments are needed to address diseases and other conditions beyond the scope of traditional Chinese medicine, which today still represents 30%-50% of treatments, according to the World Health Organization. Yet China is not simply trying to improve domestic medical care, it is also seeking to claim a profitable share of the global biopharmaceutical market.

Compared to the United States and other developed markets that currently dominate clinical development and sales of prescription drugs, China arguably has greater life sciences infrastructure and institutional support. The country's focus on technological self-sufficiency has led to more STEM graduates than the rest of the world combined and about 50% of the world's total.¹ The country produces 120,000 PhDs every year, three times the US output.²

In terms of research and development, spending as a percentage of gross domestic product (GDP) rose to around 2.7% of GDP in 2023 from 0.9% 20 years prior³ and China has approached parity with the United States in gross domestic spending on R&D (Exhibit 1). China surpassed the United States in total clinical trials in 2021 and continues to widen its lead. These advantages allow China to become a leading maker of commercial molecules to treat cancer and other diseases with large patient populations.

Exhibit 1: Chinese R&D Spending Reaching Parity with Western Economies

Source: OECD, Main Science and Technology Indicators Database, March 2025. Purchasing Power Parity (PPP) are the rates of currency conversion that aim to equalise the purchasing power of different currencies by eliminating differences in price levels between countries. GERD is the OECD's preferred measure of R&D activity for use in international comparisons. Past performance is not an indicator or a guarantee of future results.

Evolving Toward Blockbuster Drug Development

Chinese biotechnology companies are not yet at the leading edge in developing breakthrough drugs with blockbuster potential. However, they have thus far been successful either licensing and commercializing drugs from Western innovators or developing drugs that act as "fast followers" in large disease markets established by Western companies. Chinese firms often license drug candidates developed in the West from smaller US biotechnology companies that lack the capital and regulatory expertise to sell into the domestic Chinese market. In the case of fast followers, certain

technologies that become commoditized, such as antibodies and certain small-molecule drugs, can now be produced more effectively by Chinese companies.

One of the drivers of new commercial therapeutics in China is the country's National Reimbursable Drug List (NRDL)—a list of drugs for which the state health insurance system will reimburse out-of-pocket costs to a large proportion of its patients. China has a population four times the size of the United States, and a higher prevalence of medical conditions such as certain cancers. While China agrees to pay a fraction of the US price for prescription drugs, it can partially offset this with higher patient volumes. Drugs targeting oncology and rare diseases will likely command lower price concessions due to increased need. In addition, a new development of a C-class drug list (to be finalized in 2025) allows for more costly drugs to fall under China's health insurance coverage without being held to the same price concession standards of the NRDL.

The paths to profitability in the Chinese biotech industry are evolving rapidly and there will be a mix of business models, from developing products for domestic to a combination of domestic and foreign markets. Profitability at the moment is slim, but we expect to see this industry continue to evolve quickly and gain attention across the biopharmaceutical space.

Cost and efficiency are the primary selling points for Chinese labs in developing early-stage compounds for large cap partners. Over the last several years, these Chinese biotech companies have leveraged their production expertise and deep population of biologists and chemists to develop an impressive innovation engine. An expanding number of Chinese biotechs are now producing new compounds that multinational pharmaceutical firms are licensing in accelerating numbers (Exhibits 2 and 3).

Exhibit 2: Licensing Deals Outside China are Accelerating?

*2025 count is annualized projection. Data as of March 31, 2025. Source: DealForma. Licensing deals with upfront payments of US\$10 million or more. There is no assurance that any estimate, forecast or projection will be realized. Past performance is not an indicator or a guarantee of future results.

Exhibit 3: With Chinese Firms Gaining Share in Large Pharma Deals

Data as of March 29, 2025. Source: DealForma. Licensing deals with upfront payments of US\$50 million or more. There is no assurance that any estimate, forecast or projection will be realized. Past performance is not an indicator or a guarantee of future results.

The combined value of China's outside licensing deals reached around US\$46 billion in 2024, up from US\$38 billion in 2023 and US\$28 billion in 2022, according to data provider NextPharma. Meanwhile, the number of global companies licensing into China has decreased across the same period. These tailwinds have helped China expand its share of global drug development to nearly 30% compared to 48% for the United States, according to data provider Citeline. Strong IP protection has positioned China to receive global investment, with a 2024 policy encouraging more IP collaboration between global and Chinese companies. US investment bank Stifel projects that molecules licensed by large pharmaceutical firms from China will increase to 37% in 2025. This shift has been largely driven by US companies seeking cheaper drug development alternatives and has led to R&D spending in China outpacing that of the United States.

While the current trade standoff between China and the United States could impact access to Chinese companies and their IP, the financial benefits for Western pharmaceutical companies of partnering with a well-funded, well-run Chinese biotech innovation engine are clear. For Chinese drug developers, bringing original treatments to market through licensing deals and marketing partnerships creates a compelling investment case. In this fast-developing industry, we are actively looking at the key players.

Endnotes

Source: Center for Security and Emerging Technology, The Global Distribution of STEM Graduates: Which Countries Lead the Way? November 27, 2023.

Source: fDi Intelligence, "China's universities outpace US peers amid tech competition", April 30, 2025.

Source: National Bureau of Statistics of China, December 27, 2024.

WHAT ARE THE RISKS?

All investments involve risks, including possible loss of principal.

Equity securities are subject to price fluctuation and possible loss of principal.

Investments in fast-growing industries like the health care sector could result in increased price fluctuation, especially over the short term, due to the rapid pace of product change and development and changes in government regulation of companies emphasizing scientific or technological advancement or regulatory approval for new drugs and medical instruments.

International investments are subject to special risks, including currency fluctuations and social, economic and political uncertainties, which could increase volatility. These risks are magnified in emerging markets. Investments in companies in a specific country or region may experience greater volatility than those that are more broadly diversified geographically.

The government's participation in the economy is still high and, therefore, investments in China will be subject to larger regulatory risk levels compared to many other countries.

There are special risks associated with investments in China, Hong Kong and Taiwan, including less liquidity, expropriation, confiscatory taxation, international trade tensions, nationalization, and exchange control regulations and rapid inflation, all of which can negatively impact the fund. Investments in Taiwan could be adversely affected by its political and economic relationship with China.

WF: 5994664

CONTRIBUTORS

Marshall Gordon

Senior Research Analyst for Healthcare ClearBridge Investments

Elisa Mazen

Head of Global Growth Portfolio Manager ClearBridge Investments

Related Content

Explore ClearBridge Investments Insights

RELATED CONTENT

IMPORTANT LEGAL INFORMATION

This material is intended to be of general interest only and should not be construed as individual investment advice or a recommendation or solicitation to buy, sell or hold any security or to adopt any investment strategy. It does not constitute legal or tax advice. This material may not be reproduced, distributed or published without prior written permission from Franklin Templeton.

The views expressed are those of the investment manager and the comments, opinions and analyses are rendered as at publication date and may change without notice. The underlying assumptions and these views are subject to change based on market and other conditions and may differ from other portfolio managers or of the firm as a whole. The information provided in this material is not intended as a complete analysis of every material fact regarding any country, region or market. There is no assurance that any prediction, projection or forecast on the economy, stock market, bond market or the economic trends of the markets will be realized. The value of investments and the income from them can go down as well as up and you may not get back the full amount that you invested. Past performance is not necessarily indicative nor a guarantee of future performance. All investments involve risks, including possible loss of principal.

Any research and analysis contained in this material has been procured by Franklin Templeton for its own purposes and may be acted upon in that connection and, as such, is provided to you incidentally. Data from third party sources may have been used in the preparation of this material and Franklin Templeton ("FT") has not independently verified, validated or audited such data. Although information has been obtained from sources that Franklin Templeton believes to be reliable, no guarantee can be given as to its accuracy and such information may be incomplete or condensed and may be subject to change at any time without notice. The mention of any individual securities should neither constitute nor be construed as a recommendation to purchase, hold or sell any securities, and the information provided regarding

such individual securities (if any) is not a sufficient basis upon which to make an investment decision. FT accepts no liability whatsoever for any loss arising from use of this information and reliance upon the comments, opinions and analyses in the material is at the sole discretion of the user.

Franklin Templeton has environmental, social and governance (ESG) capabilities; however, not all strategies or products for a strategy consider ?ESG? as part of their investment process.

Products, services and information may not be available in all jurisdictions and are offered outside the U.S. by other FT affiliates and/or their distributors as local laws and regulation permits. Please consult your own financial professional or Franklin Templeton institutional contact for further information on availability of products and services in your jurisdiction.

Issued in the U.S. by Franklin Templeton, One Franklin Parkway, San Mateo, California 94403-1906, (800) DIAL BEN/342-5236, franklintempleton.com. Investments are not FDIC insured; may lose value; and are not bank guaranteed.

You need Adobe Acrobat Reader to view and print PDF documents. Download a free version from Adobe's website.

CFA® and Chartered Financial Analyst® are trademarks owned by CFA Institute.